

DATA HANDLING AND VISUALIZATION (DSA06)

Experiment Set 13: Geographic Data Analysis

Question / Task:

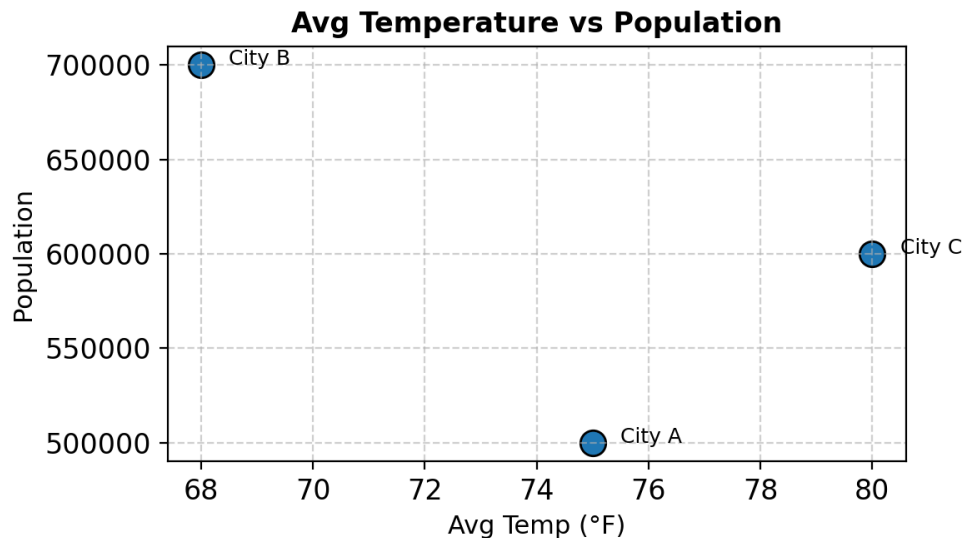
1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.
2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

R Program Code:

```
# Set 13: Geographic Data Scatter Plot (R)
city <- c("City A", "City B", "City C")
pop <- c(500000, 700000, 600000)
temp <- c(75, 68, 80)

plot(temp, pop, col = "#1F74B4", pch = 19, cex = 1.8,
      main = "Avg Temperature vs. Population Size",
      xlab = "Avg Temperature (°F)", ylab = "Population")
text(temp + 0.5, pop, labels = city, pos = 4)
grid()
```

Output Screenshot:



Experiment Set 14: Survey Results Analysis

Question / Task:

1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart

elements.

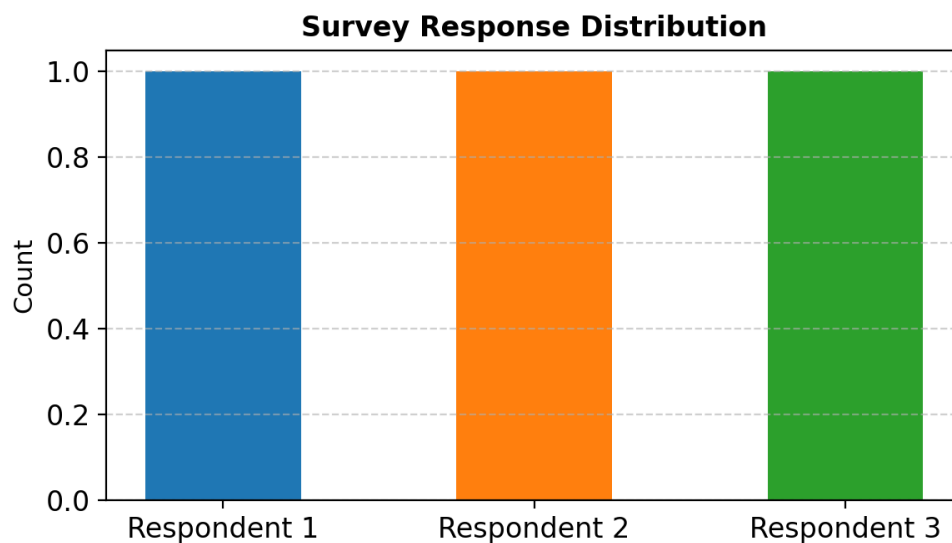
2. Generate a radar chart to visualize the overall pattern of responses across all three questions.

R Program Code:

```
# Set 14: Survey Results Distribution (R)
resp <- c("Respondent 1", "Respondent 2", "Respondent 3")
ans <- c(1, 1, 1)

barplot(ans, names.arg = resp, col = c("#1F74B4", "#FF7F0E", "#2CA02C"),
        main = "Survey Response Distribution", ylab = "Count")
grid(nx = NA, ny = NULL)
```

Output Screenshot:



Experiment Set 15: Monthly Product Sales Analysis

Question / Task:

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.
2. Generate a stacked area chart to represent overall sales trend for all products over three months.

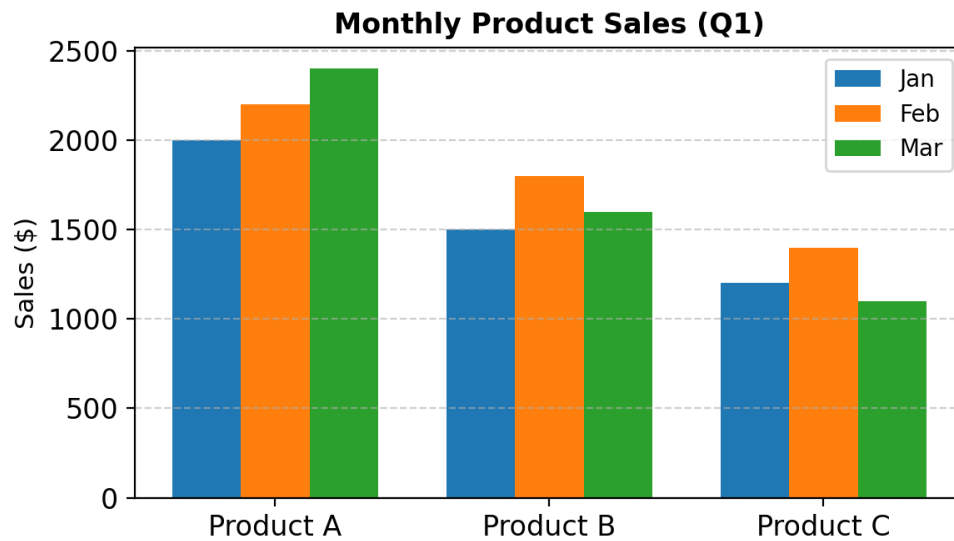
R Program Code:

```
# Set 15: Q1 Product Sales Grouped Bar Chart (R)
sales_data <- matrix(c(2000, 2200, 2400,
                      1500, 1800, 1600,
                      1200, 1400, 1100), nrow = 3, byrow = TRUE)
colnames(sales_data) <- c("Jan", "Feb", "Mar")
rownames(sales_data) <- c("Prod A", "Prod B", "Prod C")

barplot(sales_data, beside = TRUE, col = c("#1F74B4", "#FF7F0E", "#2CA02C"),
        main = "Monthly Product Sales (Q1)", ylab = "Sales ($)")
```

```
legend("topright", legend = rownames(sales_data), fill = c("#1F74B4", "#FF7F0E", "#2CA02C"))
```

Output Screenshot:



Experiment Set 16: Customer Demographics Analysis

Question / Task:

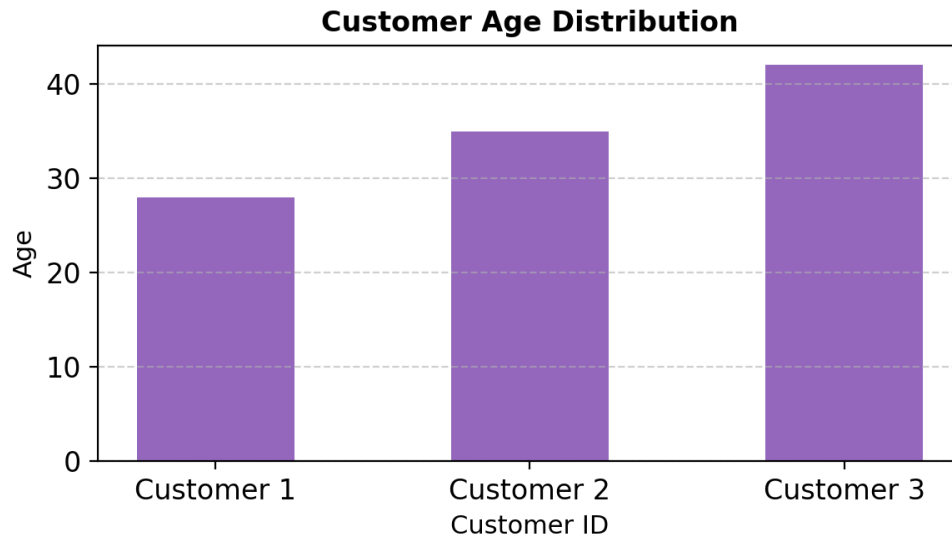
1. Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.
2. Generate a pie chart to represent the distribution of customers by gender.

R Program Code:

```
# Set 16: Customer Demographics Bar Chart (R)
cust <- c("Customer 1", "Customer 2", "Customer 3")
ages <- c(28, 35, 42)
```

```
barplot(ages, names.arg = cust, col = "#9467BD",
        main = "Customer Age Distribution",
        xlab = "Customer ID", ylab = "Age")
grid(nx = NA, ny = NULL)
```

Output Screenshot:



Experiment Set 17: Employee Performance Analysis

Question / Task:

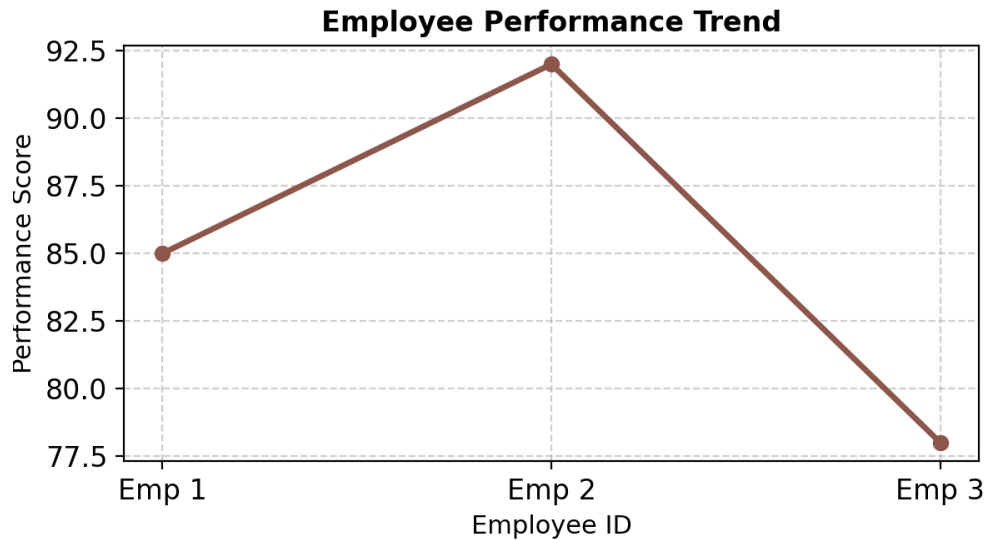
1. Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.
2. Generate a bar chart showing the distribution of employees across different departments.

R Program Code:

```
# Set 17: Employee Performance Trend Line Chart (R)
emp <- c("Emp 1", "Emp 2", "Emp 3")
score <- c(85, 92, 78)

plot(score, type = "o", col = "#8C564B", pch = 19, lwd = 2.5,
      main = "Employee Performance Trend",
      xlab = "Employee ID", ylab = "Performance Score", xaxt = "n")
axis(1, at = 1:3, labels = emp)
grid()
```

Output Screenshot:



Experiment Set 18: Product Inventory Management

Question / Task:

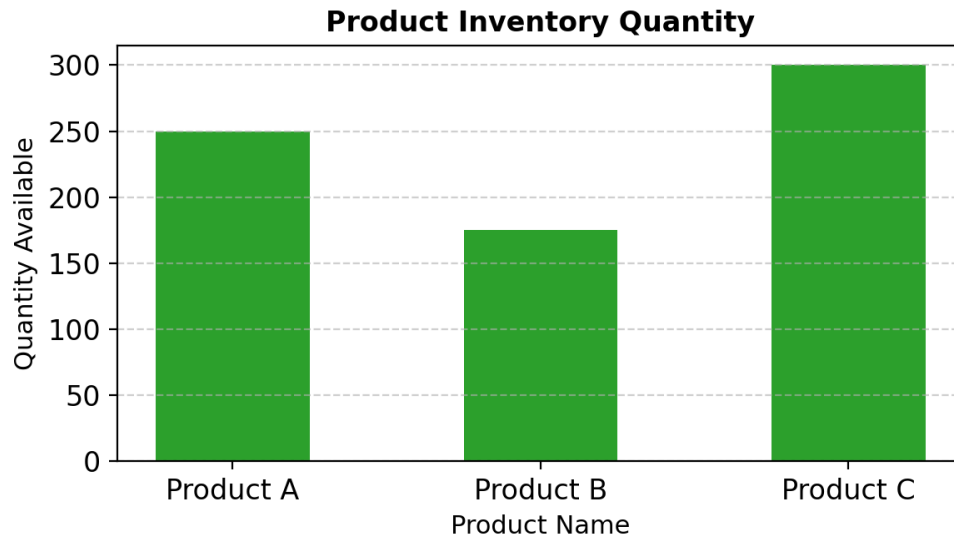
1. Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.
2. Generate a stacked bar chart to show the quantity of each product within different categories.

R Program Code:

```
# Set 18: Product Inventory Bar Chart (R)
products <- c("Product A", "Product B", "Product C")
qty <- c(250, 175, 300)

barplot(qty, names.arg = products, col = "#2CA02C",
        main = "Product Inventory Quantity",
        xlab = "Product Name", ylab = "Quantity Available")
grid(nx = NA, ny = NULL)
```

Output Screenshot:



Experiment Set 19: Survey Responses Analysis

Question / Task:

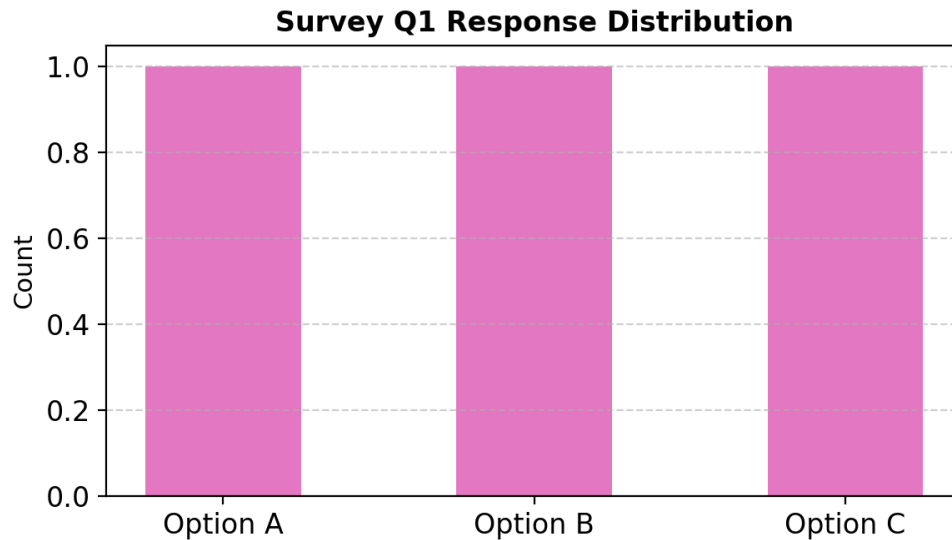
1. Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.
2. Generate a stacked bar chart to represent overall response distribution across questions.

R Program Code:

```
# Set 19: Survey Responses Distribution (R)
answers <- c("Option A", "Option B", "Option C")
counts <- c(1, 1, 1)

barplot(counts, names.arg = answers, col = "#E377C2",
        main = "Survey Q1 Response Distribution", ylab = "Count")
grid(nx = NA, ny = NULL)
```

Output Screenshot:



Experiment Set 20: Stock Analysis

Question / Task:

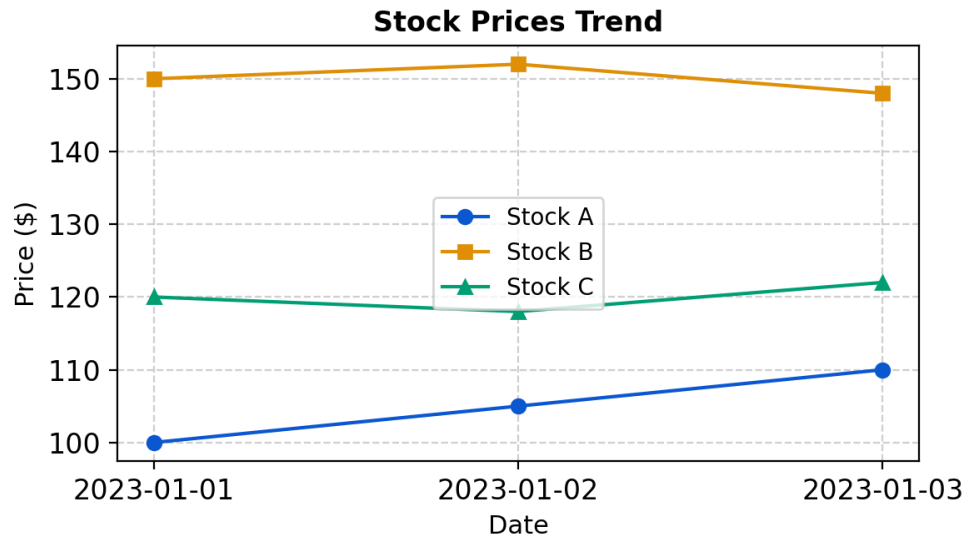
1. Create a line chart to visualize stock prices of three companies (Stock A, Stock B, Stock C) over time.
2. Generate a bar chart showing the daily percentage change in stock prices for Stock A.

R Program Code:

```
# Set 20: Stock Prices Line Chart (R)
dates <- c("2023-01-01", "2023-01-02", "2023-01-03")
stockA <- c(100, 105, 110)
stockB <- c(150, 152, 148)
stockC <- c(120, 118, 122)

plot(stockA, type = "o", col = "#1F74B4", pch = 19, ylim = c(90, 160),
      main = "Stock Prices Trend", xlab = "Date", ylab = "Price ($)", xaxt = "n")
lines(stockB, type = "o", col = "#FF7F0E", pch = 15)
lines(stockC, type = "o", col = "#2CA02C", pch = 17)
axis(1, at = 1:3, labels = dates)
legend("center", legend = c("Stock A", "Stock B", "Stock C"), col = c("#1F74B4",
"#FF7F0E", "#2CA02C"), lty = 1)
grid()
```

Output Screenshot:



Experiment Set 21: Energy Consumption Data Analysis

Question / Task:

1. Import dataset. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.
2. Create a scatter plot between Temperature and Units_Consumed.

R Program Code:

```
# Set 21: Energy Consumption Histogram (R)
units <- c(320, 540, 880, 350, 610, 920)

hist(units, col = "#17BECF", border = "white",
      main = "Energy Units Consumed Distribution",
      xlab = "Units Consumed (kWh)", ylab = "Frequency")
grid(nx = NA, ny = NULL)
```

Output Screenshot:

